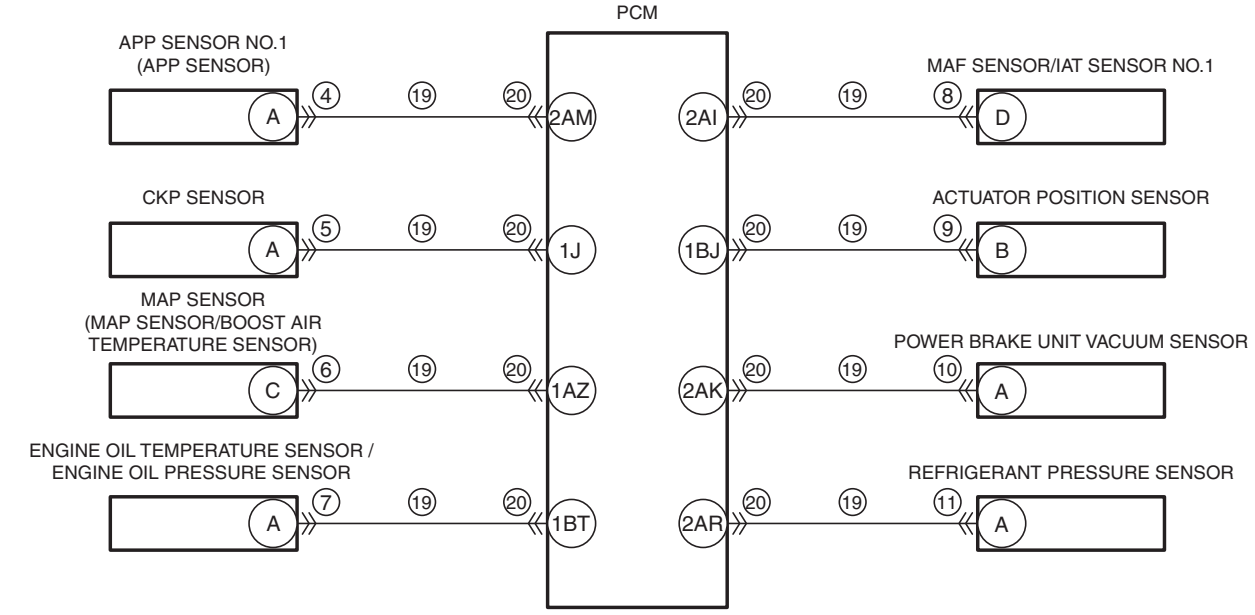
DTC P0642:00 [SKYACTIV-D 1.5]

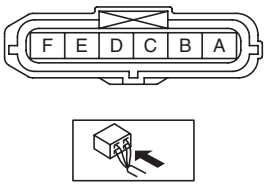
id0102q2315500

DTC P0642:00	Constant voltage power supply circuit low input
DETECTION CONDITION	• When the following condition is met, the output voltage of the **5 V** power supply terminal **4.63 V or less** for a continuous **1 s**: **MONITORING CONDITIONS** — Battery voltage: **8 V or more** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on) **Diagnostic support note** • This is a continuous monitor (CCM). • The check engine light illuminates if the PCM detects the above malfunction condition during the first drive cycle. • FREEZE FRAME DATA/Snapshot data is available. • DTC is stored in the PCM memory.
FAIL-SAFE FUNCTION	• PCM restricts engine torque. • Fully open the intake shutter valve. • Inhibits the EGR control. • Inhibits engine-stop by operating the i-stop function.
POSSIBLE CAUSE	• APP sensor connector or terminals malfunction • CKP sensor connector or terminals malfunction • MAP sensor/boost air temperature sensor connector or terminals malfunction • Engine oil temperature sensor/engine oil pressure sensor connector or terminals malfunction • MAF sensor/IAT sensor No.1 connector or terminals malfunction • Actuator position sensor connector or terminals malfunction • Power brake unit vacuum sensor connector or terminals malfunction • Refrigerant pressure sensor connector or terminals malfunction • Clutch stroke sensor connector or terminals malfunction • Exhaust shutter valve connector or terminals malfunction • Intake shutter valve connector or terminals malfunction • Exhaust gas pressure sensor No.1 connector or terminals malfunction • Exhaust gas pressure sensor No.2 connector or terminals malfunction • Coolant control valve connector or terminals malfunction • Speed sensor assy connector or terminals malfunction • Short to ground in wiring harness between the following terminals: — APP sensor terminal A—PCM terminal 2AM — CKP sensor terminal A—PCM terminal 1J — MAP sensor/boost air temperature sensor terminal C—PCM terminal 1AZ — Engine oil temperature sensor/engine oil pressure sensor terminal A—PCM terminal 1BT — MAF sensor/IAT sensor No.1 terminal D—PCM terminal 2AI — Actuator position sensor terminal B—PCM terminal 1BJ — Power brake unit vacuum sensor terminal A—PCM terminal 2AK — Refrigerant pressure sensor terminal A—PCM terminal 2AR — Clutch stroke sensor terminal C—PCM terminal 2AL — Exhaust shutter valve terminal A—PCM terminal 1AK — Intake shutter valve terminal E—PCM terminal 1AU — Exhaust gas pressure sensor No.1 terminal A—PCM terminal 1CJ — Exhaust gas pressure sensor No.2 terminal A—PCM terminal 1AA — Coolant control valve terminal F—PCM terminal 1BO — Speed sensor assy terminal D—PCM terminal 1BI • PCM connector or terminals malfunction • PCM malfunction

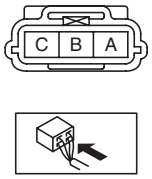
DTC P0642:00	Constant voltage power supply circuit low input



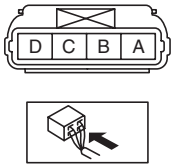
APP SENSOR WIRING HARNESS-SIDE CONNECTOR



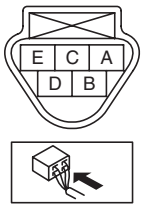
CKP SENSOR WIRING HARNESS-SIDE CONNECTOR



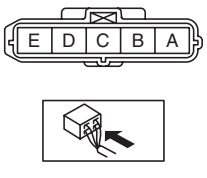
MAP SENSOR/ BOOST AIR TEMPERATURE SENSOR WIRING HARNESS-SIDE CONNECTOR



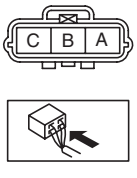
ENGINE OIL TEMPERATURE SENSOR/ ENGINE OIL PRESSURE SENSOR WIRING HARNESS-SIDE CONNECTOR



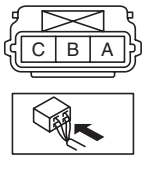
MAF SENSOR/IAT SENSOR NO.1 WIRING HARNESS-SIDE CONNECTOR



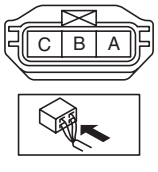
ACTUATOR POSITION SENSOR WIRING HARNESS-SIDE CONNECTOR



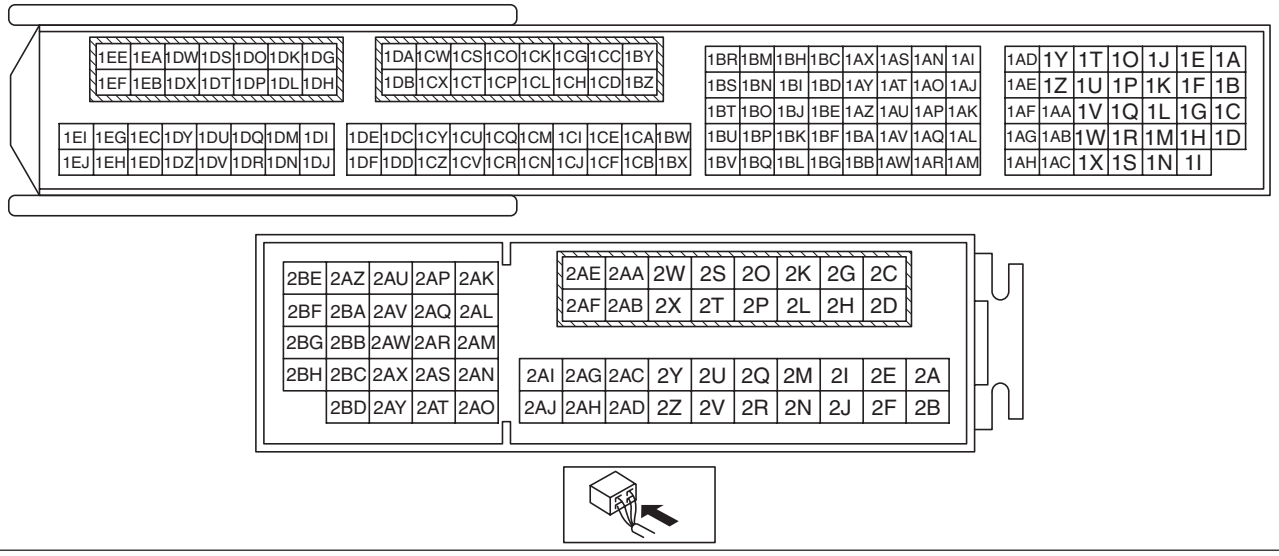
POWER BRAKE UNIT VACUUM SENSOR WIRING HARNESS-SIDE CONNECTOR



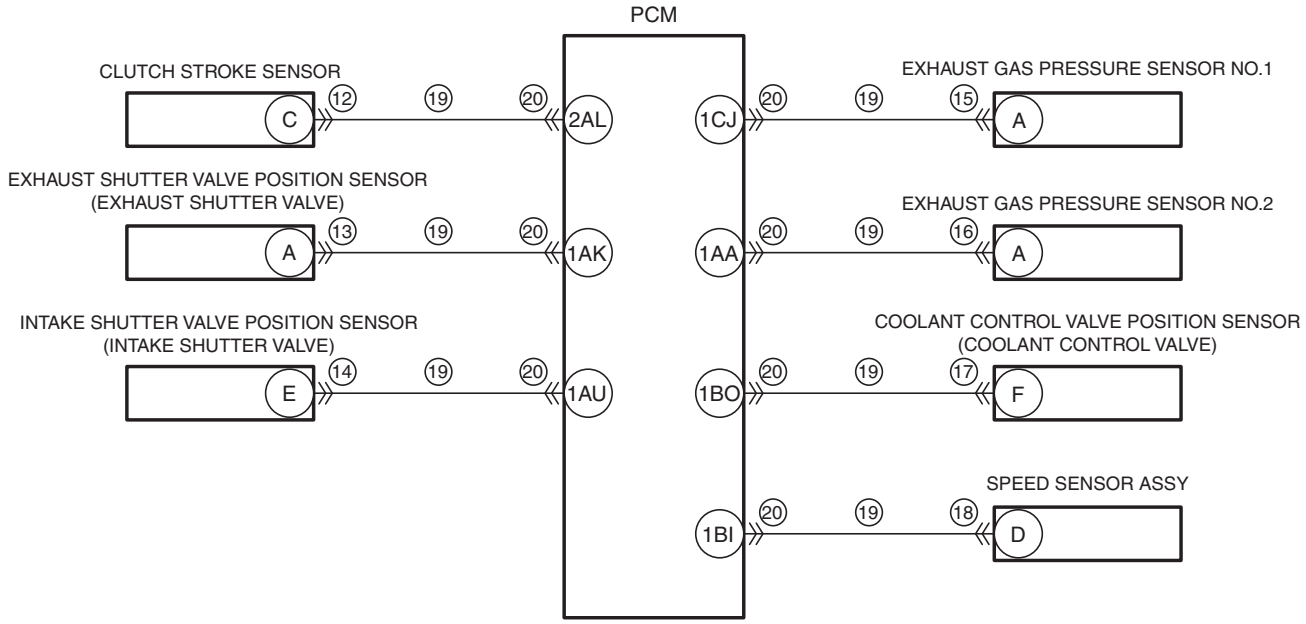
REFRIGERANT PRESSURE SENSOR WIRING HARNESS-SIDE CONNECTOR



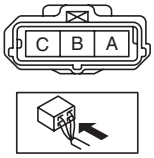
PCM WIRING HARNESS-SIDE CONNECTOR



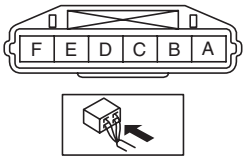
DTC P0642:00	Constant voltage power supply circuit low input



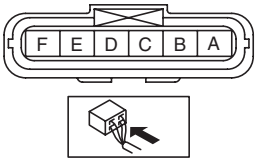
CLUTCH STROKE SENSOR WIRING HARNESS-SIDE CONNECTOR



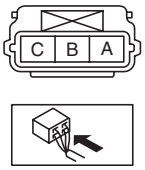
EXHAUST SHUTTER VALVE WIRING HARNESS-SIDE CONNECTOR



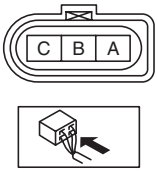
INTAKE SHUTTER VALVE WIRING HARNESS-SIDE CONNECTOR



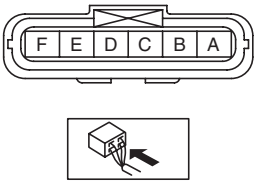
EXHAUST GAS PRESSURE SENSOR NO.1 WIRING HARNESS-SIDE CONNECTOR



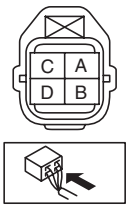
EXHAUST GAS PRESSURE SENSOR NO.2 WIRING HARNESS-SIDE CONNECTOR



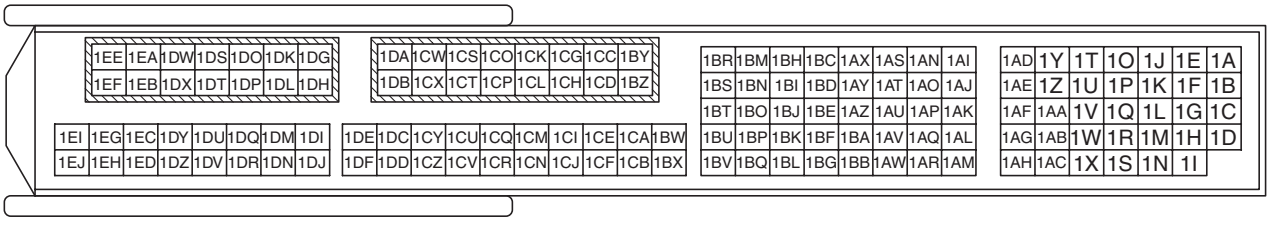
COOLANT CONTROL VALVE WIRING HARNESS-SIDE CONNECTOR

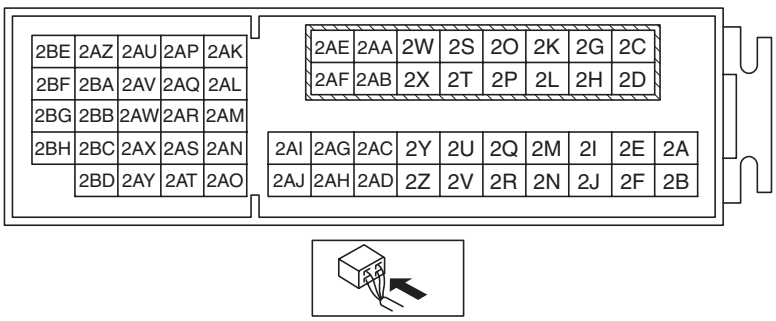


SPEED SENSOR ASSY WIRING HARNESS-SIDE CONNECTOR



PCM WIRING HARNESS-SIDE CONNECTOR





Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**RECORD VEHICLE STATUS AT TIME OF DTC DETECTION TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the FREEZE FRAME DATA/snapshot data on the repair order.	—	Go to the next step.
2	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
3	**VERIFY RELATED PENDING CODE AND/OR DTC** • Switch the ignition off, then ON (engine off). • Perform the Pending Trouble Code Access Procedure and DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Are any other PENDING CODEs and/or DTCs present?	Yes	Go to the applicable PENDING CODE or DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	Go to the next step.
4	**INSPECT APP SENSOR CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the APP sensor connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 21.
		No	Go to the next step.
5	**INSPECT CKP SENSOR CONNECTOR CONDITION** • Disconnect the CKP sensor connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 21.
		No	Go to the next step.
6	**INSPECT MAP SENSOR/BOOST AIR TEMPERATURE SENSOR CONNECTOR CONDITION** • Disconnect the MAP sensor/boost air temperature sensor connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 21.
		No	Go to the next step.
7	**INSPECT ENGINE OIL TEMPERATURE SENSOR/ENGINE OIL PRESSURE SENSOR CONNECTOR CONDITION** • Disconnect the engine oil temperature sensor/ engine oil pressure sensor connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 21.
		No	Go to the next step.
8	**INSPECT MAF SENSOR/IAT SENSOR NO.1 CONNECTOR CONDITION** • Disconnect the MAF sensor/IAT sensor No.1 connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 21.
		No	Go to the next step.

STEP	INSPECTION		ACTION
9	**INSPECT ACTUATOR POSITION SENSOR CONNECTOR CONDITION** • Disconnect the actuator position sensor connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 21.
		No	Go to the next step.
10	**INSPECT POWER BRAKE UNIT VACUUM SENSOR CONNECTOR CONDITION** • Disconnect the power brake unit vacuum sensor connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 21.
		No	Go to the next step.
11	**INSPECT REFRIGERANT PRESSURE SENSOR CONNECTOR CONDITION** • Disconnect the refrigerant pressure sensor connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 21.
		No	Go to the next step.
12	**INSPECT CLUTCH STROKE SENSOR CONNECTOR CONDITION** • Disconnect the clutch stroke sensor connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 21.
		No	Go to the next step.
13	**INSPECT EXHAUST SHUTTER VALVE CONNECTOR CONDITION** • Disconnect the exhaust shutter valve connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 21.
		No	Go to the next step.
14	**INSPECT INTAKE SHUTTER VALVE CONNECTOR CONDITION** • Disconnect the intake shutter valve connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 21.
		No	Go to the next step.
15	**INSPECT EXHAUST GAS PRESSURE SENSOR NO.1 CONNECTOR CONDITION** • Disconnect the exhaust gas pressure sensor No. 1 connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 21.
		No	Go to the next step.
16	**INSPECT EXHAUST GAS PRESSURE SENSOR NO.2 CONNECTOR CONDITION** • Disconnect the exhaust gas pressure sensor No. 2 connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 21.
		No	Go to the next step.
17	**INSPECT COOLANT CONTROL VALVE CONNECTOR CONDITION** • Disconnect the coolant control valve connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 21.
		No	Go to the next step.

STEP	INSPECTION		ACTION
18	**INSPECT SPEED SENSOR ASSY CONNECTOR CONDITION** • Disconnect the speed sensor assy connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 21.
		No	Go to the next step.
19	**INSPECT EACH POWER SUPPLY CIRCUIT FOR SHORT TO GROUND** • Verify that the following connector parts are disconnected. — APP sensor — CKP sensor — MAP sensor/boost air temperature sensor — Engine oil temperature sensor/engine oil pressure sensor — MAF sensor/IAT sensor No.1 — Actuator position sensor — Power brake unit vacuum sensor — Refrigerant pressure sensor — Clutch stroke sensor — Exhaust shutter valve — Intake shutter valve — Exhaust gas pressure sensor No.1 — Exhaust gas pressure sensor No.2 — Coolant control valve — Speed sensor assy • Inspect for continuity between the following terminals (wiring harness-side) and body ground: — APP sensor terminal A — CKP sensor terminal A — MAP sensor/boost air temperature sensor terminal C — Engine oil temperature sensor/engine oil pressure sensor terminal A — MAF sensor/IAT sensor No.1 terminal D — Actuator position sensor terminal B — Power brake unit vacuum sensor terminal A — Refrigerant pressure sensor terminal A — Clutch stroke sensor terminal C — Exhaust shutter valve terminal A — Intake shutter valve terminal E — Exhaust gas pressure sensor No.1 terminal A — Exhaust gas pressure sensor No.2 terminal A — Coolant control valve terminal F — Speed sensor assy terminal D • Is there continuity?	Yes	If the short to ground circuit could be detected in the wiring harness: • Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: — APP sensor terminal A—PCM terminal 2AM — CKP sensor terminal A—PCM terminal 1J — MAP sensor/boost air temperature sensor terminal C—PCM terminal 1AZ — Engine oil temperature sensor/engine oil pressure sensor terminal A—PCM terminal 1BT — MAF sensor/IAT sensor No.1 terminal D—PCM terminal 2AI — Actuator position sensor terminal B—PCM terminal 1BJ — Power brake unit vacuum sensor terminal A—PCM terminal 2AK — Refrigerant pressure sensor terminal A—PCM terminal 2AR — Clutch stroke sensor terminal C—PCM terminal 2AL — Exhaust shutter valve terminal A—PCM terminal 1AK — Intake shutter valve terminal E—PCM terminal 1AU — Exhaust gas pressure sensor No.1 terminal A—PCM terminal 1CJ — Exhaust gas pressure sensor No.2 terminal A—PCM terminal 1AA — Coolant control valve terminal F—PCM terminal 1BO — Speed sensor assy terminal D—PCM terminal 1BI **If there is a common connector:** — Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. — Repair or replace the malfunctioning part. **If there is no common connector:** — Repair or replace the wiring harness which has a short to ground. If the short to ground circuit could not be detected in the wiring harness: • Replace the PCM (short to ground in the PCM internal circuit). (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to Step 21.
		No	Go to the next step.
20	**INSPECT PCM CONNECTOR CONDITION** • Disconnect the PCM connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to the next step.
		No	Go to the next step.

STEP	INSPECTION		ACTION
21	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Perform the DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
22	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.